Python Foundations – Activity Handouts

Activity 0.1 – Everyday Task as an Algorithm

Title: From Daily Routine to Algorithm

Instructions to Students

1. Choose one simple daily activity (examples: making tea, preparing to go to class, booking a cab).
2. Write the steps in clear, numbered points.
3. Make sure each step is:
   - Unambiguous
   - In correct order
   - Does not skip important details

Template

1. Activity chosen: Preparing to go to class
2. Write the steps:
   1. I'll wake at 6:00AM in the morning
   2. I'll brush my teeth and wash face
   3. I'll make nesessary arrangements for class
   4. I'll iron my clothes if need.
   5. I'll take bath and get ready
   6. after having breakfast in mess i will attend class
3. Check your algorithm:
   - Can a friend follow it without asking questions? Yes ✓ / No
   - If "No", refine the steps here:
     -

Python Foundations – Activity Handouts

Activity 0.1 – Everyday Task as an Algorithm

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   - If "No", refine the steps here:
     -

Activity 0.2 – From Steps to Pseudo-code

Title: Writing Pseudo-code with IF and REPEAT

Instructions

1. Take your steps from Activity 0.1.
2. Identify where decisions happen (IF/ELSE).
3. Identify where repetition happens (REPEAT / WHILE).
4. Rewrite your algorithm using the following keywords:
   - IF, ELSE, ELSE IF
   - REPEAT, WHILE, UNTIL

Template

1. Original activity: preparing to going class
2. Pseudo-code version:

text

START

wake at 6:00 Am in morning

IF I wake up at 7:00 THEN

i will skip taking bath

ELSE

i will : skip breakfast.

ENDIF

REPEAT

_____

UNTIL _____

END

3. Reflection:
   - Where did you use IF/ELSE? _____
   - Where did you use REPEAT/WHILE? _____

Activity 1.1 – Run and Edit Given Code

Title: First Python Script – Edit and Run

Instructions

1. Your teacher will give you a small Python script (example below)
2. Do NOT write new code first; only change text or numbers
3. Run the program and observe the output.

Example Script

python

```
name = "Student"
age = 18
print("Hello,", name)
print("You are", age, "years old.")
```

Tasks

1. Change name and age to your own values.
   - name: SHREYAS PATEL
   - age: 22
2. Run the program. What is the output?

text

Output:

Hello SHREYAS PATEL
You are 22 years old.

3. Change the printed message to:
   - "Welcome to Python, <your name>"

Write your edited code here:

python

```
name = "SHREYAS PATEL"
print("Welcome to Python, <your name>"
```

output :

welcome to python SHREYAS PATEL

 # Edited code:

4. Reflection:
   - What changed in the output and why?

Printed message is change because I entered different message.

Activity 1.2 – Predict-Then-Run

Title: Predict Program Output

Instructions

1. Look at each code snippet.
2. BEFORE running, write what you think the output will be.
3. Then run it and compare.

Snippet A

python

```
x = 5
y = 2
z = x + y * 3
print(z)
```

- Predicted output: 21
- Actual output: 11

Snippet B

python

```
a = 10
b = a - 3
a = a + 1
print(a, b)
```

- Predicted output: 10,7
- Actual output: 10,9

Reflection

- Where was your prediction wrong? Why?

Snippet A because I followed BODMAS Rule

Activity 2.1 – Calculator in Steps

Title: From Word Problem to Code

Problem 1

A student has marks in 3 subjects. Find total and average.

1. Write the steps in plain language:

   1. Lik marks of 3 subjects
   2. calculate total marks from 3 subjects
   3. divide total marks by 3
   4. total and average are obtained

2. Now convert to Python (fill in the blanks):

python

mark1 = 21

mark2 = 25

mark3 = 23

total = 69

average = 23

```
print("Total =", total)
print("Average =", average)
```

3. Test with your own numbers and record output.

Activity 2.2 – Fix the Errors

Title: Debugging Simple Python

Instructions

Each snippet has an error.

1. Write what you think is wrong.
2. Correct the code.

Snippet 1

python

```
print("Hello World)
```

- Error: ”
- Corrected code: print (" Hello world.")

python

undefined

Snippet 2

python

[illegible]

[illegible]

[illegible]

[illegible]

- [illegible] bits [illegible] string [double quoted [illegible] should not be used]
- Corrected code

```
a = 10
b = 5
print ('')
```

[illegible]

[illegible]

[illegible]

- What did you learn about syntax vs logic errors?

Syntax errors does not allow program to run

logic errors run the programme but affect output

Activity 3.1 – Rule-based Classification

Title: Grades with If/Else

Rule

- If mark ≥ 90 : Grade A
- Else if mark ≥ 75 : Grade B
- Else : Grade C

Tasks

1. Draw the decision in simple text:
   - Condition 1: if mark is more than or equal to 90 -> grade A
   - Condition 2: if mark is more than or equal to 75 -> grade B
   - or else grade C
2. Now write Python code:

python

```
mark = int(input("Enter mark: "))
if mark ≥ 90 : Grade A :
  print("Grade A")
elif mark ≥ 75 : grade B :
  print("Grade B")
else:
  print("Grade C")
```

3. Test with at least 3 marks and note outputs.

---

if mark is 95
output -> grade A

mark = 81
output -> grade B

mark = 39
output -> grade C

Activity 3.2 – Mini Problem Bank

Title: Structured Problem Solving with IF/ELSE

For each question, fill the template.

Template

1. Restate the problem in one sentence:

   Sowing of seeds based on rainfall prediction (PP)

2. Conditions needed (bullets):

   - if rain comes sow the seeds (PP > 500mm)
   - else no sowing of seeds

3. Python code:

python

 # Your solution

4. Test cases and results:

   - Input: _____ Output: _____
   - Input: _____ Output: _____

Use this template for:

- Odd or even number.
- Ticket discount if age < 12 or > 60.
- Check if a year is a leap year (simple rule).

---

1) number = int(input("enter a number")
   if number, % 2 = 0:
   print ("Even")
   else:
   print ("odd")

2) age = int(input ("Enter age"))
   if age < 12 or > 60: (i.e 12 < age < 60:
   Print ("Discount in ticket")
   else
   Print("No discount in ticket")

3) Check if a year is a leap year.
   year = int(input ("Enter year"))
   if year. % 4 = 0:
   print ("year is leap year")
   else:
   print("Not leapyear")

Activity 4.1 – Sum with Loops

Title: Sum from 1 to N

Steps

1. In plain language:
   - What is "sum from 1 to N"? 1+2+3+4 ... +N
2. Write the algorithm:

1. write compute (N+1)
2. then multiply N with (N+1)
3. then divide the product with 2.
4. then the quotient value is sum of number

3. Now code it:

python

```
N = int(input("Enter N: "))
total = 0
for i in range( 1,N ):
  total = total + i
print("Sum =", total)
```

4. Try N = 5, 10, 100 and check results

i) N = 5

$$sum = \frac{5(6)}{2} = \frac{30}{2} = 15$$

For N = 10

$$sum = \frac{10 \times 11}{2} = \frac{110}{2} = 55$$

For N = 100

$$sum = \frac{100 \times 101}{2} = \frac{10100}{2} = 5050$$

Activity 4.2 – Star Patterns

Title: Loops for Patterns

Pattern (triangle)

```
*
**
***
****
```

1. Explain the pattern in words:

   The pattern is triangle

2. Write code:

python

```
rows = int(input("Enter number of rows: "))
for i in range(1, rows + 1):
    print("*" * i)
```

3. Try different rows and observe.

```
row = int (input("enter number of rows:")
columns = int(input("enter number of columns:")
for i in range.(3)
    for j in range.(3)
        print (" *", end=" ")
    print ()
```

Activity 5.1 – Working with Lists

Title: Analyze Marks List

Given

python

marks = [56, 78, 90, 45, 88]

1. In words, write how to find:
   - Maximum mark: ~~90~~ max than any other in list.
   - Average mark: total marks/5
   - Count of marks ≥ 60: which is max than @ equal to 60

2. Now write Python:

```
python

marks = [56, 78, 90, 45, 88]


# Max

max_mark = 90


# Average

total = 0

for m in marks:

  total = 56+78 +90+45+88

average = total
          5

# Count >= 60

count = 0

for m in marks:

  if count >= 60

    count = 3


print("Max:", max_mark)

print("Average:", average)

print("Count >= 60:", count)
```

Activity 5.2 – Choosing an Approach

Title: Which Method is Better?

Problem

Given a list of 1000 numbers, find how many are positive.

1. Approach A (loop) – describe in words:

for loop

2. Approach B (using built-in functions or list comprehension) – describe:

if else function if no. is >0 is +ve or -ve

3. Which is easier to understand for you now? Why?

built in function is easy to understand

4. Implement one approach in Python:

python

numbers = [10, -3, 5, 0, 8, -2] *# example*

 # Your code

```
number =int (input ("enter number"))
    if number. > 0;
    Print (" number is +ve")
else ;
    print ("number is -ve")
```

Activity 6.1 – Creating a Function

Title: Wrap Logic into a Function

Task

Turn average-of-marks logic into a function.

1. In words: what does the function do?

2. Write code:

python

```
def average_marks(marks_list):
  total = 0
  for m in marks_list:
    total = total + marks(m)
  avg = total // len mark List
  return avg

data = [60, 70, 80]
print("Average =", average_marks(data))
```

3. Test with another list and note output.

```
data = [10,20,15]
Print ("Average =", Average- marks (data))
Out Put;
        Average =15
def average - marks (marks _List);
    total = 0
for m in marks _List;
    total = total +m.
  if len (marks _List) >0;
  avg = total // len marks _List
      return avg.
  else:
        return =0.
```

Activity 6.2 – Problem-Solving Template

Title: Standard Template for Any Problem

Use this for any new problem.

1. Problem understanding

   - What is given (inputs)?

     Age is given.

   - What is required (outputs)?

     Wether discount is applicable or not.

   - Any constraints (limits, ranges)?

     b/w 12-60 yrs of age no discount.

2. Analysis

   - Write steps in bullets (algorithm):

     - if age is <12 50% discount.
     - if age is >60 30% discount.
     - else no discount.

3. Design

   - Will you use loops? Which? NO
   - Will you use conditions (if/else)? Yes
   - Will you use functions? Which ones? No
   - What data structures (list, etc.)? No

4. Code (Python)

python

 # Your implementation

5. Test

   - Test case 1: input 11 → output Discount applicable
   - Test case 2: input 53 → output No discount
   - Edge case: input 63 → output discount applicable.

Activity 7.1 – Capstone Problem

Title: Simple Fee Calculator

Problem

A class charges a base fee.

- If student is below 12 years, 50% discount.
- If above 60, 30% discount.
- Otherwise, no discount.

Use the full template from Activity 6.2 here.

(Attach a blank copy of the template and ask students to fill all five sections.)

```
age = int(input ("enter age"))
    if age <12:
        print ("50% discount")
    elif age >60:
        print ("30% discount")
    else:
        print ("No discount")
```

Activity 7.2 – Reflection Sheet

Title: How I Solve a New Problem

1. When I see a new problem, my first step is:

Problem understanding → check input, output dimens
& ranges if in case prob

2. Before coding, I make sure I know:

   - Inputs: _____
   - Outputs: _____

3. The tools I think about (loops, if, functions) are:

loops - for, while, if else data structure list, set, dictionary

4. One thing I will do differently from now on:

clearly understand the program to be solved